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Figure 12.8 CT Scan

Computed Tomographic (CT) Scanning

Computed tomography is originally known as computed axial tomography (CAT or CT Scan). The word tomography is derived from the Greek word *tomos* means slice and *graphe* means to write. It is a medical imaging technology employing tomography, were digital geometry processing is used to generate a three dimensional image of the internals of an object from a large series of two dimensional X ray images taken around a single axis of rotation (Figure12.8).

CT produces volumes of data which can be manipulated through a process known as windowing in order to demonstrate various structures based on their ability to block the X ray beam.

Clinical significance

- Gives a clear image of bone, soft tissues and blood vessels
- Helps in the diagnosis of injuries of the inner ears and sinuses
- To detect cancer, heart and lung disorders
- For diagnosis of spinal problems and skeletal injuries
- Helps to measure bone mineral density
- To detect stroke causing clots and hemorrhage in the brain.

Positron Emission Tomographic Scanning (PET)

PET is also computerized imaging technique unlike CT. Positron emission tomography (PET) is a nuclear medicine procedure based on the measurement